

Mod 2 contd

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$$R \subseteq A \times A$$

- Reflexive: $(x, x) \in R$, for each $x \in A$
 $\curvearrowright$
- Symmetric: $(x, y) \in R \Rightarrow (y, x) \in R$
 $\text{for every } x, y \in A$ $\longleftrightarrow$
- Transitive: $(x, y), (y, z) \in R \Rightarrow (x, z) \in R$
 +
- Antisymmetric: $(x, y), (y, x) \in R \Rightarrow x = y$
 A
- Equivalence relation

R, S, T

- Partial order: R, T, A

Hasse diagrams

$$P(A) := \{X : X \subseteq A\}$$

• Family of subsets of A

$$X, Y \subseteq A \cdot X \cup Y$$

$$x \in \cup \mathcal{C}$$

$$\mathcal{C} = \{x_1, x_2 \dots x_n \dots\}$$

$\mathcal{C}$ is a partition of A, provided

$$\cup \mathcal{C} = A : \mathcal{C} \text{ 'covers' } A$$



$$C \in \mathcal{C} \Rightarrow C \neq \emptyset$$

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$$C, D \in \mathcal{C}, C \neq D \Rightarrow C \cap D = \emptyset$$

Prop. Every partition $\mathcal{C}$ of A induces an equivalence relation on A .
Conversely, every equivalence relation R on A induces a partition of A .